EE3A Rivo Computer Organization HW3

1. $t0=0x12345678, $t1=0xc3c3c3c3
   (a) shift right can only be used for unsigned numbers
   $t1= 11000011110000111100001111000011(B)
   slr => $t2= 0001 1000 0111 1000 0111 1000 0111 1000(B)= 410548344(D)
   $t0 =0001 0010 0011 0100 0101 0110 0111 1000(B)
   $t2=00010000001100000101000001111000(B)=10305078(H) => $t2=0x10305078
   -100(D)=11111111111111111111111110011100(B)
   $t2 =00101010101011001100111011110000(B)
   $t3 =11111111111111111111111111111100(B)= 0xFFFFFFFC(H)
   (b) andi $t5, $t0, 2096128
   srl $t5, $t5, 10
2. add $t0, $s1, $zero #i=b
   Loop1:
   slt $t2, $t0, $s0 #break if $t3=0
   beq $t2, $zero Exit #break 1st for
   add $t1, $t0, $zero #j=i
   Loop2:
   slti $t2, $t1, 0 #break if $t3=1
   bne $t2, $zero, Exit1 #break 2nd for
   sll $t2, $t1, 2
   sub $t2, $s0, $t3 #t3=a-4j
   sll $t3, $t0, 2
   add $t3, $t3, $s2 #$t2=address of U[i]
   sw $t2, 0($t3) #U[i]=a-4j
   addi $t1, $t1, -1
   j Loop2
   Exit1:
   addi $t0, $t0, 1 #i++
   j Loop1
   Exit:
3. $t0=-52(D) => $t1=0 => $t0= 11111111111111111111111111001100
   12 = 00000000000000000000000000001100 => $t2=$t0=0xFFFFFFCC
4. PC=0x10000000
   (a) Possible. PC can be set to 0x1a000000 right before the instruction “j label” is executed, where label is at 0x1a000000.
   (b) Possible. 0fffa000(H)= 268410880(D), 10000000(H)=268435456(D), the difference is within 16 bits.

5. $t1=i, $s1=MyCounter, $s2=YourArray

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MyCounter=1
for(i=Mycounter*4; i<100; i+=4){
        MyCounter+=YourArray[i/4]*16
}
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6. $t0~7=8~15, $t8~9=24~25, $s0~7=16~23
   (1) 9
   (2) 19
   (3) 3
   (4) 19
   (5) 18
   (6) 4
   (7) 24
   (8) 9
   (9) 8
   (10) 7061